

Interpret limits of accuracy and rounding

Solve problems involving limits of accuracy, estimates and rounding in real-life contexts.

Step 58

★ Today I am learning to interpret limits of accuracy and rounding.

1. Rounding to a Given Number of Decimal Places

Round each number to the given number of decimal places.

Think about it!

- Look at the digit in the place value just after the one you are rounding to.
- 0, 1, 2, 3, 4 → round down.
- 5, 6, 7, 8, 9 → round up.

- a) 23.4567 to 1 decimal place = _____
- b) 19.842 to 2 decimal places = _____
- c) 0.98765 to 3 decimal places = _____
- d) 74.3951 to 1 decimal place = _____

- e) 5.67891 to 2 decimal places = _____
- f) 0.04567 to 3 decimal places = _____
- g) 123.45678 to 2 decimal places = _____
- h) 9.9954 to 3 decimal places = _____

2. Rounding to a Given Significant Figures

Round each number to the given number of significant figures.

Think about it!

- Count the non-zero digits from left to right (zeros at the end are not significant).
- Apply the same rule: 0–4 round down, 5–9 round up.

- a) 0.0037856 to 2 significant figures = _____
- b) 45.678 to 3 significant figures = _____
- c) 1234.567 to 4 significant figures = _____
- d) 0.01996 to 2 significant figures = _____

- e) 9876.5 to 3 significant figures = _____
- f) 0.0004567 to 2 significant figures = _____
- g) 56.7893 to 4 significant figures = _____
- h) 2.995×10^3 to 3 significant figures = _____

3. Limits of Accuracy

Write each measurement to the correct degree of accuracy.

Remember:

- If a value is correct to the nearest unit, the actual value is within $\pm \frac{1}{2}$ of that unit.
- e.g. correct to the nearest 0.1 → ± 0.05
- Lower bound = value $- \frac{1}{2}$ unit
- Upper bound = value $+ \frac{1}{2}$ unit

- a) A length is 12.36 cm correct to the nearest 0.01 cm.
Write the lower and upper bounds.
Lower bound: _____ cm Upper bound: _____ cm
- b) A mass is 2.5 kg correct to the nearest 0.1 kg.
Lower bound: _____ kg Upper bound: _____ kg
- c) A temperature is 18 °C correct to the nearest 1 °C.
Lower bound: _____ °C Upper bound: _____ °C
- d) A capacity is 3.27 L correct to the nearest 0.01 L.
Lower bound: _____ L Upper bound: _____ L

- e) A distance is 125 m correct to the nearest 10 m.
Lower bound: _____ m Upper bound: _____ m
- f) A weight is 4.32 g correct to the nearest 0.01 g.
Lower bound: _____ g Upper bound: _____ g
- g) A speed is 67.8 km/h correct to the nearest 0.1 km/h.
Lower bound: _____ km/h Upper bound: _____ km/h
- h) A time is 36 s correct to the nearest 5 s.
Lower bound: _____ s Upper bound: _____ s

4. Calculations and Rounding

Calculate each answer. Give your final answer rounded as directed.

Think about it!

- Do the calculation on your calculator first.
- Round your final answer, not each step.

- a) $23.456 + 7.891 =$ _____ (to 2 d.p.)
- b) $19.87 - 4.326 =$ _____ (to 2 d.p.)
- c) $3.456 \times 1.89 =$ _____ (to 3 s.f.)
- d) $45.67 \div 2.3 =$ _____ (to 2 d.p.)

- e) $12.345 + 67.891 - 9.876 =$ _____ (to 2 d.p.)
- f) $(2.67 \times 3.8) + 1.234 =$ _____ (to 3 s.f.)
- g) $78.56 \div 4.7 =$ _____ (to 2 d.p.)
- h) $0.00456 \times 129.8 =$ _____ (to 3 s.f.)

5. Real-World Problems

Solve each problem. Give your answer to the degree of accuracy stated.

Think about it!

- Read carefully and identify the accuracy or rounding stated.
- Use limits of accuracy to find greater or smaller possible values.

- a) A runner completes a race in 45.678 seconds.
The time is recorded to the nearest 0.01 second.
Find the lower and upper bounds for the time.
Lower bound: _____ s Upper bound: _____ s
- b) A petrol pump shows 23.478 litres in a car's tank.
The reading is correct to 3 significant figures.
Write the value of the petrol to the nearest litre.
Answer: _____ litres

- c) A shop buys 248.567 kg of rice.
The mass is correct to the nearest 0.1 kg.
What is the greatest possible mass of the rice?
Answer: _____ kg
- d) The length of a garden is 15.2 m correct to the nearest 0.5 m and the width is 7.8 m correct to the nearest 0.5 m. Find the greatest possible area of the garden.
Answer: _____ m²

6. Estimation and Checking

Answer each question.

Remember:

- Estimation helps check if your answer is sensible.
- Always include the correct units in your final answer.

- a) Estimate the answer to 18.6×4.9 by rounding to 1 significant figure first.

- b) The exact answer to a calculation is 36.987.
A student rounds the answer to 37.0 correct to 1 decimal place.
Is this reasonable? Explain your answer.

- c) A measurement is given as 56.3 cm correct to the nearest 0.1 cm.
Someone writes it as 56.35 cm.
Is this appropriate? Explain your answer.

- d) A calculation gives 0.004872.
Round this to 1 significant figure and explain what the result means in the context of the number.

Answer: _____

Answer: _____

Answer: _____

Answer: _____

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more practice.



I need help.

Parent Observation (optional)

Please tick the boxes that apply.

- ☐ Understands concepts ☐ Explains thinking clearly
- ☐ Uses number skills accurately ☐ Needs more practice
- ☐ Solves problems confidently

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____